Topic 17B: Carbonyl compounds
6. be able to identify the aldehyde and ketone functional groups
7. understand that aldehydes and ketones: i do not form intermolecular hydrogen bonds and this affects their physical properties ii can form hydrogen bonds with water and this affects their solubility

Students should:
8. understand the reactions of carbonyl compounds with: i Fehling’s or Benedict’s solution, Tollens’ reagent and acidified dichromate(VI) ions *In equations, the oxidising agent can be represented as [O]* ii lithium tetrahydridoaluminate (lithium aluminium hydride) in dry ether *In equations, the reducing agent can be represented as [H]* iii HCN, in the presence of KCN, as a nucleophilic addition reaction, using curly arrows, relevant lone pairs, dipoles and evidence of optical activity to show the mechanism iv 2,4-dinitrophenylhydrazine (2,4-DNPH), as a qualitative test for the presence of a carbonyl group and to identify a carbonyl compound given data for the melting temperatures of derivatives *The equation for this reaction is not required* v iodine in the presence of alkali